

Software Requirements Specification

for

ThermoShell

Version 0 (Draft)

Prepared by

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Revision History

Name	Date	Reason For Changes	Version
ThermoWare	9/19/26	Initial SRS draft for assesment	0
ThermoWare	9/29/26	Added in-text IEEE citations	0.1
ThermoWare	9/30/26	Updated draft for submission	0.2

1. Introduction

1.1 Purpose

This document specifies the software and system requirements for ThermoShell and its companion mobile/web application, ThermoShell Connect. It covers the initial prototype release (Version 0.1) developed by Team ThermoWare for the CS-480 Capstone Project Design course at WVU Institute of Technology, under the advisement of Dr. Omer Lateef. This SRS covers the full system: vest hardware and firmware and the companion app.

1.2 Document Conventions

Functional requirements are denoted by the prefix REQ followed by an abbreviation of the feature and further identified by a sequence number. Each requirement is assigned a priority of High, Medium, or Low. Items still under team discussion are given a “[TBD]” tag and collected in Appendix C so they can be decided upon later.

1.3 Intended Audience and Reading Suggestions

This document is intended for the ThermoWare team, our faculty advisor Dr. Lateef, and Dr. Waseem and Dr. Zahmeeth for assessment. Those new to the project should start with Sections 1–2 for a high-level overview. For those already familiar with the project, jump to Section 4 for the features. Please note that Section 5 covers safety and other nonfunctional constraints that apply across all subsystems.

1.4 Product Scope

ThermoShell is a wearable vest for winter sports athletes (primarily skiers and snowboarders) that combines zone-based automatic body heating with biometric and activity tracking. Heating zones across the torso use body-temperature sensors to regulate warmth automatically, while heart-rate and GPS sensors track the wearer's vitals, speed, distance, and run/lap data. All of this is viewable, controllable, and shareable through the ThermoShell Connect app. The product is designed to be fully washable, with all electronics and sensors removable, and to support multi-day use via hot-swappable batteries. While primarily for winter sports, the same zone-heating and tracking capability can apply to other cold-weather use cases such as construction, and outdoor repair work.

1.5 References

- [1] A. M. Stoll and L. C. Greene, “Relationship between pain and tissue damage due to thermal radiation,” *J. Appl. Physiol.*, vol. 14, no. 3, pp. 373–382, May 1959.
- [2] A. R. Moritz and F. C. Henriques, “Studies of thermal injury: II. The relative importance of time and surface temperature in the causation of cutaneous burns,” *Am. J. Pathol.*, vol. 23, no. 5, pp. 695–720, Sep. 1947.

[3] Raspberry Pi Ltd, “Raspberry Pi Zero 2 W product brief,” Raspberry Pi Ltd, Cambridge, U.K., Apr. 2024. [Online]. Available: <https://datasheets.raspberrypi.com/rpizero2/raspberry-pi-zero-2-w-product-brief.pdf>. [Accessed: Sep. 22, 2026].

[4] Arduino S.r.l., “Arduino® Nano 33 BLE Sense Rev2 datasheet,” ABX00069/ABX00070, Arduino, Monza, Italy, 2023. [Online]. Available: <https://docs.arduino.cc/resources/datasheets/ABX00069-datasheet.pdf>. [Accessed: Sep. 22, 2026].

2. Overall Description

2.1 Product Perspective

ThermoShell is a self-contained product that consists of two interconnected subsystems. The first is the vest hardware, built around a SBC that reads zone temperature, heart-rate, and GPS sensors and drives the copper-ribbon heating elements. Second is the ThermoShell Connect companion app (phone and web), which communicates with the vest over Bluetooth to display biometric data, allow manual heating control, and let users view and share activity history. The vest is designed to regulate zone heating automatically even when not connected to the app; the app adds monitoring, manual control, and data logging on top of that autonomous behavior.

2.2 Product Functions

- Sense body temperature at each defined torso zone.
- Automatically activate/deactivate each zone's heating element to maintain a target temperature.
- Continuously monitor and log heart rate.
- Track GPS position, speed, distance, and user-marked run/lap times.
- Sync biometric and activity data to the ThermoShell Connect app over Bluetooth.
- Allow manual override of zone heating from the app.
- Support full detachment of all electronics, sensors, and the battery so the vest can be washed.
- Support hot-swapping batteries for extended use.

2.3 User Classes and Characteristics

ThermoShell is primarily meant for winter sport athletes like skiers and snowboarders, as well as hiker and runners. Our secondary user class is construction, road, and maintenance crews. Our tertiary audience are casual users who dislike cold temperatures in the winter. The device must be fully operable through gloves and a clothing layer over the vest.

2.4 Operating Environment

- SBC: Candidates are the Raspberry Pi Zero 2 W running a lightweight operating system like Raspberry Pi OS Lite [3] and the Arduino Nano running in the Arduino Sketch environment [4] [TBD]. Regardless of the unit we use, it will be operating outdoors in winter conditions including cold temperatures, snow, and moisture exposure.

- Companion app: Android plus a web dashboard accessible from a standard Chromium or Firefox based browser.
- Connectivity: Bluetooth and/or 2.4GHz Wi-Fi [TBD].

2.5 Design and Implementation Constraints

- Budget: soft cap of \$500 for prototype parts. Current estimate is roughly \$350.
- Schedule: system design finalized and all parts on hand by December 1st. Assembly, integration, app development and bug fixing, and the final demonstration occur next semester, with a goal of having an assembled vest and a working app ready to test in cold weather months early in the spring term, and full completion by early May 2027.
- All electronics, sensors, and the battery must connect via removable/quick-disconnect interfaces rather than being permanently wired, to support washability.
- Battery form factor and connector must support hot-swapping.
- Wireless communication limited to Bluetooth and/or 2.4GHz Wi-Fi.

2.6 User Documentation

ThermoWare has not finalized documentation. A quick-start guide covering charging, washing, and app pairing is planned [TBD].

2.7 Assumptions and Dependencies

- Assumes selected hardware components remain available and within budget through the current semester's purchasing window.
- Assumes continued meetings with Dr. Lateef to discuss, review and get feedback throughout both the Fall 2026 and Spring 2027 semesters.
- Assumes the course timeline (parts in-hand by December, final demo by May) does not shift.

3. External Interface Requirements

3.1 User Interfaces

The user interface is contained within ThermoShell Connect, which provides a real-time body temperature, heart rate, and GPS data, as well as manual controls to override heating function, and finally an activity history for viewing and sharing past activities. The vest itself is not currently planned to include an onboard display, however, a minimal set of onboard controls for overriding heating and tracking activity, with indicators for power/temperature/connection status is planned [TBD].

3.2 Hardware Interfaces

- Raspberry Pi Zero 2 W: quad-core Linux SBC, interfacing over GPIO/I²C/SPI/UART with the temperature sensors, heating-element, heart-rate sensor, and GPS module [3]. More processing power and native Python/Linux tooling for handling several simultaneous sensor reads plus Bluetooth and Wi-Fi [TBD].

- Arduino Nano: Compact, and has a built-in BLE 5.0 radio and an onboard 9-axis IMU [4]. Interfaces with external I²C/SPI/UART sensors [TBD].
- Removable electronics: sensors and the SBC connect to the fabric via a quick-disconnect connector [TBD], allowing full removal for washing.
- Battery management system: a swappable connector supporting hot-swap replacement without shutdown. Would mean having an internal battery to support device long enough for swap to occur.

3.3 Software Interfaces

- SBC logic: dependent on the hardware choice. If the Raspberry Pi Zero 2 W, then the best option will be Python running on Raspberry Pi OS, or if the Arduino Nano is chosen, then C/C++ on Arduino sketch firmware should be used.
- ThermoShell Connect data transfer: over Bluetooth and/or Wi-Fi [TBD].
- App data storage: how we will store heart rate and GPS data before and after syncing with ThermoShell Connect [TBD].

3.4 Communication Interfaces

- Bluetooth: relatively short-range pairing and allows for control of the vest from the ThermoShell Connect app.
- 2.4GHz Wi-Fi: higher-bandwidth sync when a network is available; better for large amounts of data like GPS.
- Transfer data encryption [TBD].

4. System Features

4.1 Zone Heating and Temperature Regulation

4.1.1 Description and Priority

Automatically regulates temperature using multiple heating zones across the torso. Each zone has its own temperature sensor and copper-ribbon heating element and targets a warmer temperature without user input. This is a core feature that separates ThermoShell from other heated clothing on the market. Zone heating and temperature regulation is a high-priority feature.

4.1.2 Stimulus/Response Sequences

1. User sets a target temperature for a zone or general body temperature via the app.
2. The vest continuously reads each zone's temperature sensor.
3. If any of the zone's temperature readings is below the target threshold, the system activates that zone's heating element.
4. If any of the zone's temperature reaches the target threshold, the system deactivates that zone's heating element [1], [2].

If a zone sensor reading exceeds the maximum safe temperature, the system forces a shut off and notifies the user.

4.1.3 Functional Requirements

REQ-ZH-1	The system should read temperature from each zone's sensor at a defined polling interval [interval TBD, e.g. every 2 seconds].	High Priority
REQ-ZH-2	The system should activate a zone's heating element when that zone's temperature falls below its configured target threshold.	High Priority
REQ-ZH-3	The system should deactivate a zone's heating element when that zone's temperature reaches its configured target threshold.	High Priority
REQ-ZH-4	Each heating zone should be controlled independently of all other zones.	High Priority
REQ-ZH-5	The user will be able to set a target temperature per zone through the ThermoShell Connect app.	High Priority
REQ-ZH-6	The system will view the maximum safe skin-contact temperature as a hard cutoff, independent of the target-temperature logic [1], [2].	High Priority

4.2 Heart Rate Monitoring

4.2.1 Description and Priority

Continuously monitors the user's heart rate during use and makes that data viewable in real time and as session history in the ThermoShell Connect app. Heart rate monitoring is a medium-priority feature

4.2.2 Stimulus/Response Sequences

1. The heart rate sensor begins continuous measurement.
2. Heart-rate readings are timestamped and logged to the SBC's storage device.
3. When connected to the app, heart rate data offloads and syncs to the app in near real time.

4.2.3 Functional Requirements

REQ-HR-1	The system should continuously measure heart rate whenever the vest on.	Medium Priority
REQ-HR-2	The system should log heart rate readings with timestamps to the SBC's storage device when not connected to ThermoShell Connect.	Medium Priority
REQ-HR-3	The system should sync logged heart rate history to ThermoShell Connect whenever possible.	Medium Priority
REQ-HR-4	The system should alert the user to unusual heart rate readings [TBD].	Medium Priority

4.3 GPS and Activity Tracking

4.3.1 Description and Priority

Tracks the wearer's location, speed, distance, and lap times during activities, giving athletes performance data to see how their speed and run times change. GPS and activity tracking is a high-priority feature.

4.3.2 Stimulus/Response Sequences

1. GPS acquires the wearer's position at a regular interval.
2. System computes current speed and cumulative distance from subsequent GPS pings.
3. User marks the start and end of a run from the app [TBD].
4. System logs a run/lap summary with info like duration, distance, average speed, max speed and can sync it to ThermoShell Connect.

4.3.3 Functional Requirements

REQ-GPS-1	The system will sample GPS position in timed intervals [TBD].	High Priority
REQ-GPS-2	The system should derive current speed and total distance traveled from GPS data.	High Priority
REQ-GPS-3	The user should be able to set start and end points of a run via ThermoShell Connect.	Medium Priority
REQ-GPS-4	The system should log each lap's duration, distance, and speed summary to the app.	Medium Priority

4.4 Removable Electronics

4.4.1 Description and Priority

All electronic components, including sensors, the SBC and the battery must be removable so the fabric of the vest can be washed. Removable electronics is a high-priority feature

4.4.2 Stimulus/Response Sequences

1. User disconnects the SBC, battery, and all sensors from the vest. Heating ribbon is not removed.
2. Fabric vest is washed independently of any electronics and hung out to dry.
3. User reattaches the SBC, battery, and sensors.
4. On next boot, the system performs a connection check across all sensors and heating elements.

4.4.3 Functional Requirements

REQ-RE-1	All electronic components should be removable from the vest without the need for tools [TBD].	High Priority
REQ-RE-2	The battery should be removable from the vest independently of the other components.	High Priority
REQ-RE-3	All temperature, heart-rate, and GPS sensors should be detachable.	High Priority
REQ-RE-4	The system should perform a diagnostic check for loose connections and report results to ThermoShell Connect.	Medium Priority

4.5 Hot-Swappable Battery

4.5.1 Description and Priority

Supports replacing the battery without losing system state, so users can extend a day's use with a spare battery. Hot-swappable Batteries is a medium-priority feature

4.5.2 Stimulus/Response Sequences

1. Battery charge drops to a low-battery threshold.
2. ThermoShell Connect notifies the user via the app (and/or an onboard indicator, TBD).
3. User swaps in a spare battery.
4. System resumes normal operation without a full reboot, retaining zone-heating settings and the current session's logged data.

4.5.3 Functional Requirements

REQ-HSB-1	A single fully-charged battery should power the vest for a certain amount of hours [TBD].	Medium Priority
REQ-HSB-2	The system should support hot-swapping the battery without losing the zone heating setting or the current session's heart rate or GPS data.	High Priority
REQ-HSB-3	The system should have the current battery level viewable on ThermoShell Connect.	Medium Priority
REQ-HSB-4	When the system reaches the low-battery threshold, the user should be notified	Medium Priority

5. Other Nonfunctional Requirements

5.1 Performance Requirements

- Temperature sensors should be read on a timed interval to conserve power [TBD].
- GPS position should also be read on a timed interval to conserve power. This will need to be balanced with accuracy of the location data [TBD].
- The ThermoShell Connect app should reflect live vest data in real time or with low latency, and do the same for controlling the heating element.

5.2 Safety Requirements

- The maximum long-term allowable skin-contact temperature for any heating zone should be 109°F according to research by Stoll and Greene (1959) and Moritz and Henriques (1947) [1], [2].
- The battery pack will have short-circuit and overheating protection due to the moist environment it will be used in.

5.3 Security Requirements

- The ThermoShell Connect app will require user authentication.

- Storing and transferring data encryption [TBD].
- Heart-rate and GPS location being sensitive data, will require a data sharing agreement between the user and the service [TBD].

5.4 Software Quality Attributes

- Reliability: Vest should have consistent power draw, the software should be built.
- Usability: The vest must be operable through gloves and an outer layer, have an easily understandable UX and have removable electronics.
- Durability: The vest material should hold up when being washed and all other components should withstand possible impacts while in use.
- Maintainability: Vest logic and app code will go through multiple cycles of refactoring for ease of maintenance later on.

5.5 Business Rules

Since ThermoShell is in the product planning stage, it is simply too early to make considerations for business rules at this point.

6. Other Requirements

Diagrams, reports, and other forms of documentation are requirements of CS-480, but don't fall into hardware and software requirements given in this document.

Appendix A: Glossary

Zone	A controlled heating region on the vest.
Heating Element	A copper-ribbon element that produces heat via resisting current
SBC	Single Board Computer
GPS	Global Positioning System
ThermoShell Connect	The companion mobile and web app for ThermoShell
Hot-swap	Replacing the battery pack while the system remains operational.
UX	User Experience

Appendix B: Analysis Models

ThermoWare plans to add data flow, class, and entity-relationship diagrams [TBD].

Appendix C: To Be Determined List

1. Final data-gathering and control SBC: Raspberry Pi Zero 2 W vs. Arduino Nano (Sections 2.4, 3.2, 3.3).
2. Which subsystems should use Bluetooth vs. Wi-Fi (Sections 2.4, 3.2, 3.3, 3.4).
3. Quick-disconnect connector type for the removable electronics harness (Section 3.2).

4. Onboard control and indicator design, including override buttons and power/temperature/connection status indicators (Section 3.1).
5. Local data storage for heart-rate and GPS data before and after syncing to app (Section 3.3).
6. Data encryption (Sections 3.4, 5.3).
7. Data sharing and privacy agreement terms for heart-rate and GPS location data (Section 5.3).
8. Sensor and GPS ping intervals, accuracy, and power draw (Sections 4.1.3 REQ-ZH-1, 4.3.3 REQ-GPS-1, 5.1).
9. Abnormal heart rate alert thresholds and behavior (Section 4.2.3, REQ-HR-4).
10. Run/lap start control: app button, onboard button, or automatic detection (Section 4.3.2).
11. Tool requirement for removing electronics (Section 4.4.3, REQ-RE-1).
12. Determine battery life (Section 4.5.3, REQ-HSB-1).
13. End-user documentation format and content (Section 2.6).
14. Analysis diagrams for Appendix B: data flow, class, and entity-relationship diagrams.